A-A-50304
25 March 1991

COMMERCIAL ITEM DESCRIPTION

DYNAMOMETER, ENGINE RUN-IN, 500 HORSEPOWER CAPACITY

The General Services Administration has authorized the use of this commercial item description in preference to Military Specification MIL-D-45561C(AL) dated 26 May 1987.

Abstract. This description covers a dynamometer designed and constructed to facilitate the testing and run-in of reciprocating-piston, internal combustion engines having outputs up to and including 500 brake horsepower (HP).

Salient characteristics.

1. Design of the dynamometer shall be a contractor's commercial model capable of mounting, testing, and run-in of Department of Defense (DOD) reciprocating piston, spark ignition and compression ignition, internal combustion engines up to and including 500 horsepower at speeds from 400 to 5,000 revolutions per minute (rpm). The engine dynamometer shall be so designed that the engines may be readily installed with only minor adjustment for the particular engine being tested. The dynamometer shall include all the equipment necessary, including engine adapters, for the mounting and proper operation of the following engines: Continental AOSI-895-5, R-6602, LDS427-2 and LDS465-1; Cummins NTC-400, 6CTA, NHC-250 and 8V-300; Detroit Diesel 8V92TA, 8V-71-T, 3-53, 6V-53T and 6.2 liter; GMC302; Mack Endt-673 and IH DT 817G, except for the following, which (if required) shall be furnished by the Government:

   Coolant supply, coolant disposal, compressed air supply electric power supply, engine fuel supply, engine air inlet, engine exhaust disposal, engine hoisting equipment, fire extinguishing equipment, instruments for measuring atmospheric conditions, and storage batteries.

: Beneficial comments (recommendations, additions, deletions, clarifications and any data which may improve this document should be sent by letter to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300. :

AMSC N/A FSC 4910

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a. Lubrication shall be provided means for all bearings (except sealed-for-life type). The supplier shall include and install all required lubricants, fluids, greases and other compounds in the dynamometer.

b. Safety provisions that incorporate safety devices to prevent conditions hazardous to personnel or detrimental to equipment shall be provided including an overspeed trip governor that overrides the throttle control, and either stops the engine completely or returns the engine to idle, in the event the engine speed exceeds the governor setting. The overspeed governor shall be graduated in increments not greater than 100 rpm, and shall be adjustable through a 2,000 to 5,000 rpm engine speed range. The governor speed selector shall be panel mounted on the control console.

c. Parts shall be replaceable and adjustable without modification of the part or the dynamometer.

2. Product description. The dynamometer shall consist of:

a. Power absorption system with:

(1) Power absorber capable of continuously absorbing the following horsepower values at the revolutions per minute (rpm) without exceeding a water consumption rate of 4.5 gallons per horsepower per hour.

20HP	at	600 RPM	± 50 RPM
160HP	at	1200 RPM	± 100 RPM
335HP	at	1800 RPM	± 100 RPM
500HP	at	2300 RPM	± 100 RPM
200HP	at	5000 RPM	± 100 RPM

The minimum controllable HP curve shall not exceed the following:

10HP	at	2000 RPM	± 100 RPM
140HP	at	5000 RPM	± 100 RPM

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Power absorbers of either hydraulic, eddy current, or electric generator type with bi-directional rotation are acceptable.

(2) Power absorber frame designed for at least 20,000 hours of operation without failure or degradation of dynamometer accuracy and a frame that provides support for the power absorber.

(3) Dynamometer cooling temperature controls with:

(a) Automatic temperature controls with manual override adequate for maximum rated power absorption.

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(b) Low water pressure safety switch that is connected to the trip governor control.

(c) Replaceable screen of 20 mesh that is incorporated in the fluid line capable of successful operation with a water supply pressure ranging from 35 pounds per square inch gage (psig) to 70 psig and a water supply temperature ranging from 80 degrees F to 100 degrees F.

(4) Load controls that:

(a) Control the load on the engine to within ± one percent of the torque indicator full scale value, and ± one percent of the horsepower indicator full scale value.

(b) Provide coarse and fine adjustment of any load within the dynamometer capacity range in no more than one minute.

(c) Mount on the control console.

(5) Throttle control that:

(a) Provides throttle control for all engines listed herein.
(b) Operates in conjunction with the overspeed trip governor.
(c) Installs on the power absorber frame and connects to the engine throttle lever with an adjustable linkage.
(d) Is controlled by the control transmitter mounted on the control console.

(6) Engine fuel system that:

(a) Is supplied as part of the dynamometer system, that conducts fuel from a supply outlet, provided by the Government, through a shutoff valve, fuel filters, selector valves, fuel flow measuring units, modulating valves, and fuel lines.
(b) Is mounted on a panel which can be easily attached and removed from the power absorber frame.
(c) Has valve stem packing material that is compatible with gasoline and diesel fuel.
(d) Has all metal piping made of brass or copper or stainless steel.
(e) Permits passage of 400 pounds of fuel per hour with a pressure drop in the system of less than 4 psig.
(f) Has convenient and efficient bleed-offs provided to eliminate vapor locks.
(g) Has filters of the replaceable element type having a rating of not more than 5-microns.

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(h) Has two independent fuel systems, each with a primer pump, a shutoff and pressure modulating valve, and a pressure gage, for (1) gasoline fueled carburetor and gasoline fuel injection type engines and, (2) compression-ignition type engines.

(7) Engine electrical system that:

(a) Is self-contained unit which can be easily attached and removed from the power absorber frame.
(b) Includes provisions for two 12 volt vehicle wet storage batteries (wired in series) and an insulated wiring system for negative grounded 12 and 24 volt ignition systems.
(c) Includes provisions for remote control operation of starting relays, and booster coils at the control console.
(d) Has starting relays and circuits capable of handling maximum starting currents.
(e) Has cables conforming to NEMA Publication WC3 capable of transmitting imposed electrical currents and voltages.
(f) Has an automatic battery charger, with a two stage charging rate, equipped with an automatic regulator that provides a charging rate of 5 to 6 amperes per hour when the battery voltage drops below normal, and to provide a "trickle" charge of not more than one ampere per hour, and equipped with a meter and a meter switch.

(8) Junction box conforming to type 12 electrical enclosure per NEMA 250 attached to the power absorber frame for connecting leads to their respective system.

b. Two mobile engine stands that:

(1) Are capable of supporting engines weighing up to 5,000 pounds.
(2) Have adjustable front and rear mounts and include mounting adapters for the installation and alignment of all engines listed herein.
(3) Are mounted on a minimum of four wheels that are:

(a) Rigidly mounted non-swivel type on one end and 360 degree swivel caster type on the other end.
(b) A minimum diameter of four inches and provide free steerable movement of the loaded engine stand in any direction over the floor surface.
(c) Provided with antifriction bearings.
(d) Located on the engine stand such that the wheels effectively support movement and prevent an unbalanced or tipping condition of the stand when loaded with the maximum weight of 5,000 pounds.
(e) Provided with foot actuated brakes (swivel wheels only) to prevent movement of the engine stand.

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(4) Have mechanical guides and a locking devices that provide aligning and locking the mobile engine stands to the power absorber frame.
(5) Are equipped with a drain pan having a tapered pipe thread of sufficient size (3/4 NPT minimum) with mating pipe cap.

c. A thermostatically controlled cooling column for use with liquid cooled engines with:

(1) A heat dissipation rate of at least 1,320,000 BTU per hour.
(2) An adjustable thermostatic control valve to obtain an engine water outlet temperature of 160 degrees F to 200 degrees F.
(3) A pressure relief valve incorporated in the cooling system to prevent abnormal cooling system pressure.
(4) A means provided for filling, venting and draining the cooling system
(5) A free standing and self-contained design.
(6) Hoses with quick disconnect couplings and adapters provided to connect the cooling column with the engines listed herein.
(7) A sight gage to indicate the coolant level and a dial thermometer to indicate the engine coolant temperature.

d. Control console that:

(1) Houses all controls and instruments necessary for engine testing.
(2) Incorporates a data desk in a position below and in front of the instrumentation and control boards.
(3) Has instruments and controls oriented to promote legibility, and ease of operation by the seated operator.
(4) Is free standing and completely enclosed.
(5) Has all internal electrical leads, thermocouple leads and mechanical leads terminating on a junction panel in the rear of the control console.
(6) Has access from the rear of the control console provided for connecting electrical and mechanical leads and for servicing and repairing all instruments and controls.
(7) Is arranged for installation outside of the test cell.
(8) Has instruments and controls permanently marked for function, and grouped, as far as possible, in the four general areas of engine controls, engine instrumentation, dynamometer instrumentation and controls, and safety warning devices.
(9) Has a receptacle furnished and mounted on the control console to house a supply of test-log forms.
(10) Has a supply of 100 test-log forms.

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(11) Includes:

(a) Engine controls with:

i. Panel mounted ignition switch.
ii. Panel mounted switch for the battery system, and the engine throttle control transmitter.
iii. Start-stop controls for the priming pumps for both fuel systems.
iv. End of test shutdown actuator operated from a panel mounted stop button on the control console.
v. An air intake strangulation valve to operate from a panel mounted emergency stop button on the control console.
vi. The capability that when the engine stop button is pressed the shutdown actuator will return the fuel lever to the off position.
vii. The capability that when the emergency stop button is pressed the air intake strangulation valve will close.

(b) Engine instrumentation with the following panel mounted indicators that are capable of receiving, and indicating oil pressure, intake manifold pressure and vacuum, fuel pressure, fuel flow rate, cylinder head temperature, coolant temperature, and oil temperature as noted:

i. One mechanical engine oil pressure indicating device with a range from 0 to 160 pounds per square inch (psi) graduated in increments not greater than 5 psi, and an accuracy error not greater than + three percent of the full scale value from 0 to 40 psi and 120 to 160 psi and ± two percent of full scale value from 40 to 120 psi.
ii. Two individual or one compound fuel pressure indicating devices with one for gasoline and one for diesel fuel, a range from 0 to 15 psig pressure and 0 to 30 inches of mercury vacuum. Read out in increments not greater than .5 psi and 1 inch, and an accuracy error not greater than ± five percent of full scale value. Fuel pressure transmitted to the indicator electrically from a transducer or other means. No hard plumbing of fuel lines is permitted.
iii. One manifold pressure indicating device with a range of 0 to 70 inches of mercury absolute pressure, graduated in increments not greater than .5 inch, and an accuracy error not greater than ± 1/2 inch of mercury.
iv. One thermocouple type engine temperature indicator having at least eight stations with automatic cold junction temperature compensation, a range of 0 to 600 degrees F, and an accuracy error not greater than ± one percent of full scale value. Eight pairs of No. 14 AWG, or larger, Iron-Constantan thermocouple lead wires, special tolerance, insulated, stranded conductor, in accordance with ANSI Standard MC 96.1 for use from the instrument

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to the dynamometer junction panel thermocouple outlets. Instruments compensated to match the length of lead wire from the indicating instrument to the engine thermocouples. Eight Iron-Constantan thermocouples with compression fittings provided to mount in various locations on the engines to be tested.

v. Two fuel flow indicating devices or one compound fuel flow indicating device (one for spark-ignition fuel, and one for compression-ignition fuel) with an accuracy error not greater than ± two percent of actual value over the specified range. Working range of the device that is from 15 pounds per hour to 500 pounds per hour. A selector switch that permits the use of either spark ignition fuel or compression ignition fuel. Selector switch shall operate in conjunction with a solenoid selector valve.

(12) Has dynamometer instrumentation and control area that include the following panel mounted indicators capable of receiving, transmitting, and indicating engine speed, torque, and horsepower, as noted; digitally displaying indicators for (a) and (b) that have an accuracy error not greater than ± one percent of actual value over the specified range and ± one digit; display devices that are readable at a distance of five feet by the operator and speed, torque and horse power indicators that are equipped with built-in calibration reference for easy front-panel calibration:

(a) One magnetic pulse type engine rpm sensor with leads to an electric tachometer indicator with digital display indicator having a range from 0 to 5,000 rpm.
(b) One torque and one horsepower measuring system with torque sensing strain gage load cells directly actuated by the power absorber torque arm; with necessary leads to attach the torque sensing device to the torque and hp indicators; with torque indicator having a range from 0 to 1200 foot-pounds (minimum); and with the horsepower indicator having a range from 0 to 500 hp.
(c) One coolant supply pressure indicating device with a range of 0 to 100 psi graduated in increments not greater than 5 psi and an accuracy error within ± three percent of full scale value.
(d) One power absorber fluid outlet temperature device (liquid cooled power absorbers only) with an accuracy error within ± three percent of full scale value.
(e) Dynamometer load controls as specified herein.

(13) Safety warning devices furnished to indicate:

(a) High power absorber fluid temperature (water cooled and hydraulic type power absorbers only).
(b) Low engine oil pressure.
(c) High engine oil temperature.

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(d) High engine coolant temperature.
(e) Tripped overspeed governor.

(14) Control console electrical and mechanical leads required for remote engine operation passing through separate conduits, and grouped as follows:

(a) Instrument electrical leads
(b) Engine and operating 24 volt lead
(c) Thermocouple leads
(d) Primary power leads
(e) Mechanical leads

With the length of the leads being 30 feet (minimum). Electrical leads terminating in terminal eyes. Instrument leads with soldered connections at both ends. All leads plainly identified at both ends.

e. A dynamometer drive coupling capable of transmitting power over the horsepower and speed ranges specified herein; capable of compensating for angular misalignment of at least three degrees between the power absorber shaft and the engine shaft and the drive coupling having a lengthwise adjustment of at least 1 1/2 inches.
f. Adapter kits that provide for testing and run-in of the engines specified herein.

3. Hardware and special tools necessary for the proper installation, operation, calibration, and maintenance of the dynamometer and instruments shall be furnished.

4. Electrical components, starters, relays, switches, wiring, and their enclosures shall conform to and be located in accordance with NEMA Electrical Standards including:

a. Electric motors conforming to the requirements of NEMA MG-1. Starters conforming to the requirements of MIL-S-12514.
b. Power source designed for operation on nominal 220/440 volt, 50/60 hertz (Hz), three-phase current. Control switches and circuits operable on a nominal 115-volt, 50/60 Hz, single phase current.

5. Fungus control. Components of electrical circuitry that are hermetically sealed or inherently resistant to fungus growth and moisture need not be treated. Components of electrical circuitry that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MIL-T-152 and MIL-V-173.

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6. Identification marking. Each dynamometer shall be marked for identification as specified in MIL-STD-130 including the National Stock Number.

7. Each dynamometer to be furnished with a copy of a video tape cassette(s) which provide, by example, the setup, function and operation of the dynamometer system. The tape(s) to show as a minimum the major components of the dynamometer system. The tapes to show the complete operation of mounting and testing of at least one engine utilizing the respective adapter kit. The tape(s) to be full motion video, with audio explanatory, of personnel using the dynamometer and its components from start of operation to completion. The video cassette tape to be 3/4 inch U-Matic, color, with audio, capable of portable playing.

Contractor Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric Products. Products manufactured to metric dimensions shall be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified using conversion tables contained in Fed-Std-376, and all other requirements of this Commercial Item Description are met.

Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

Packaging. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Notes.

The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this commercial item description to the extent specified herein.

MIL-T-152	- Treatment Moisture and Fungus Resistant, Of Communication, Electronic, and Associated Electrical Equipment
MIL-V-173	- Varnish, moisture and fungus resistant (for treatment of communications, electronics and associated equipment
MIL-S-12514	- Starter, Motor; Across-the-line and Reduced Voltage Types, A-C Induction, 3/4 to 100 HP, Manual and Electric
FED-STD-H28	- Screw Threads Standards for Federal Services

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MIL-STD-130	- Identification Marking of US Military Property
MIL-STD-889	- Dissimilar Metals
NEMA Pub WC3	- Rubber Insulated Wire and Cable for the
NEMA MG-1	- Motors and Generators
NEMA 250	- Enclosures for Electrical Equipment (1000 volts max).

(Copies of specifications and standards required by the contractor in connection with specific acquisition functions should be obtained from the contracting officer.)

(Application for copies of NEMA standards should be addressed to National Electrical Manufacturer's Association, 155 East 44th Street, New York, NY 10017.)

As the result of a market acceptability survey, the following companies have shown to be manufacturers of an acceptable product described within this Commercial Item Description. This list does not limit or exclude the selection of other manufacturers who have an acceptable product provided that they meet the market acceptability survey criteria, as verified by the CID developer, that has been established for the CID. Other manufacturers' products will be considered for a current procurement if market acceptability qualification can be established before the specified date of contract award; otherwise other manufacturers' products will be considered for future solicitations.

Kahn Industries (Model 301-160) and Schenck Pegasus Corp (Part No. D700-2E (Hydraulic) OR W400 (Eddy Current)) manufacturers/suppliers.

The following is a list of contractors who have had successful contracts with the Government providing the subject item:

Superflow Corp.	Contract DAAA09-90-C-0262
Go-Power Corp.	Contract DAAA09-87-C-0749
Kahn Industries.	Contract DAAA09-88-C-0175
Clayton Industries	Contract DAAA09-80-C-5094

MILITARY INTERESTS:

Custodian
Army - AL

Review activity
Army - ME

User activity
Army - AT

CIVIL AGENCY COORDINATION ACTIVITY:
GSA - FSS (7FXE)

PREPARING ACTIVITY:
Army- AL

Project No. 4910-0688

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

CID A-A-50304 Dynamometer, Engine Run-In, 500 Horsepower Capacity

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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